

Gauge fixing flow for GFFA

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Abstract

We propose to use a gauge-fixing field transformation to fix field configurations to Landau gauge. The resulting configurations can be generated using the gauge-fixed Fourier acceleration (GFFA) [1]. This strategy avoids the inner integral over gauge transformations which is typically required to fix the gauge without Gribov copies, and as such makes the GFFA an exact algorithm with which to generate gauge field configurations.

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1 Introduction

We consider the GFFA project of Norman, Yikai and Rana [1]. This strategy is very successful in aligning the perturbative Fourier modes, but it suffers from an unacceptable overhead/inexactness resulting from an integral over all gauge transformations which is required for gauge-fixing. The current strategy is to put the log of this integral into the action. For the algorithm to be exact, it must be evaluated before and after each trajectory. Remarkably, these evaluations actually look exactly like correlation functions which can be evaluated using inner Monte Carlos (IMC). Regardless, this is an inexact solution leading to an inexact algorithm. The IMC integrand is also poorly suited to importance sampling. If one could do away with the IMC or dramatically increase sampling efficiency, the algorithm would be greatly improved.

This document starts with a rather verbose description of the issue before getting to the main idea, which is given in Section 4.

2 Gauge fixing

2.1 Gribov copies

The standard PI for YM theory

$$\int [dA] \exp \left(- S_{YM}[A] \right), \quad (1)$$

diverges in the continuum limit since the integration measure $[dA]$ is the product of an infinite number of finite-volume $SU(3)$ groups, implying a divergent volume integral. The introduction of a gauge fixing term will suppress gauge transformations that don't satisfy the gauge fixing condition and the FP determinant accounts for the local volume of the gauge orbits.

This process suffers from the so-called Gribov ambiguity, where we can actually have multiple gauge-equivalent fields satisfying the gauge condition [2]. This means that our path integral is over counting; this could, in principle, invalidate the gauge-fixing procedure. Gribov managed to show that the ambiguity is actually a result of the (near-)zero modes of the FP operator, and as such the problem doesn't actually effect perturbative calculations, see Appendix A. The problem remains in the IR, however, so this is something we have to worry about for non-perturbative calculations.

2.2 Gauge fixing without Gribov copies

First I'll describe the gauge fixing procedure used in the GFFA. We modify the correlation function as follows:

$$\begin{aligned}\langle O \rangle &= \int [dU] e^{S[U]} O[U] \\ &= \int [dU] e^{S[U]} O[U] \frac{\int [dg] X[U^g]}{I[U]} \\ &= \int [dg] \int [dU^g] e^{S[U^g]} O[U^g] X[U^g] \frac{1}{I[U^g]},\end{aligned}\tag{2}$$

where in the third line we took advantage of the fact that the action and observable are gauge-invariant so we can make a gauge transformation to $U \rightarrow U' = U^g$. We'll prove the gauge invariance of $I[U]$ shortly. Relabeling now $U^g \rightarrow U$ we find that those terms are independent of g , so we have

$$\left[\int [dg] \right] \int [dU] e^{S[U]} O[U] X[U] \frac{1}{I[U]}.\tag{3}$$

Any compact group manifold has finite volume which can be normalized to one, so that the integral of g out front is solved and removed. The remaining bit

$$I[U] = \int [dg] X[U^g],\tag{4}$$

is a integral over all gauge transformations, so making a gauge transformation $U \rightarrow U^h$ in this function doesn't change the integral. This follows from the invariance of the Haar measure under right and left translations.

For $X[U]$, we follows Refs. [3, 4]. It can be shown that the global minima of the following function exists on each gauge orbit. The fact that there is a unique global minimum allows us to avoid the Gribov ambiguity problem. Furthermore, this function is minimized for fields A satisfying Landau gauge (see appendix B)

$$X[A] = \frac{\beta M^2}{2} |A|^2.\tag{5}$$

This arises from the lattice as

$$X_{\text{lat}}[U] = \exp \left(-\frac{\beta M^2}{2} \text{ReTr} U \right),\tag{6}$$

since in the continuum limit $U = \exp(aA^b T_b)$, where a is the lattice spacing.

The integral in the denominator is thus

$$\int [dg] \exp \left(-\frac{\beta M^2}{2} |A^g|^2 \right).\tag{7}$$

If we choose the absolute minimum of this integral to occur at $g = h(A)$ and set $A_0 = A^h$, we can evaluate this as a Gaussian integral:

$$I[A] = \exp \left(-\frac{\beta M^2}{2} |A_0|^2 \right) \left[\det (\partial \cdot D(A_0)) \right]^{-1/2}.\tag{8}$$

Here we've used $g = \exp(w)$ and $A^g = A + D(A)\omega$. The path integral is

$$Z = \int [dA] \exp \left(-S_{cl}(A) - \frac{\beta M^2}{2} [|A|^2 - |A_0|^2] \right) \sqrt{\det (\partial \cdot D(A_0))}.\tag{9}$$

Now we have to consider the fact that the FP determinant actually encodes the volume of the local gauge orbit, so that under the GTF $A \rightarrow A_0^\phi = \phi^\dagger A_0 \phi + \phi^\dagger \partial \phi$, we find

$$DA = DA_0 D\phi \det(\partial \cdot D(A_0)). \quad (10)$$

Considering infinitesimal GTF $\phi = \exp(\omega)$, $A_0^\phi = A + D(A)\omega$. The cancellation of terms containing $[A, \omega]$ occurs because in the inner product we trace over these antisymmetric matrices; the details can be found in Ref. [3]. Keeping the correct ones you find

$$\left|A_0^\phi\right|^2 = |A_0|^2 - 2(\omega, \partial \cdot A_0) + (\omega, \partial \cdot D(A_0)\omega). \quad (11)$$

A_0 satisfy Landau gauge, so the middle bit vanishes. We can perform the Gaussian integral over $d\phi = d\omega + O(\omega^2)$. The result is

$$Z = \int dA_0 \exp\left(-S_{cl}\right) \det(\partial \cdot D(A_0)), \quad (12)$$

with A_0 satisfying Landau gauge.

3 Gauge fixing for GFFA

For the lattice we use Eq. (6):

$$S_{GF}(U) = \frac{-\beta M^2}{2} \sum_{x,\mu} \text{ReTr} U(x, \mu). \quad (13)$$

The we place the log of Eq. (7) into the action, and so we are left with

$$S = S_W + S_{GF} + S_{GF2}, \quad (14)$$

$$S_{GF2}(U) = \ln \int \prod_x dg(x) \exp\left(\frac{-\beta M^2}{2} \sum_{y,\mu} \text{ReTr}[g(y)U(y, \mu)g(y + \hat{\mu}^\dagger)]\right). \quad (15)$$

This must be evaluated at the beginning and end of each trajectory in order to do an accept/reject step. It is simple to show that the difference of logs results in a Monte Carlo integral

$$\ln \left\langle \exp\left(S_{GF}[U^g] - S_{GF}[U'^g]\right) \right\rangle_g. \quad (16)$$

It probably suffices to solve the integral in the limit $M \rightarrow \infty$. In continuum, this would allow us to use a Gaussian approx.

4 Gauge-fixing field transformation

In this section we describe an exact method by which we can Fourier accelerate the HMC without introducing the problematic integral over gauge transformations. To do this we introduce a field-transforming flow for the configuration U . Evolving this flow for a time t will suppress the gauge degrees of freedom in a manner similar to including Eq. (6) in the action. The resulting field configuration V is softly fixed to Landau gauge, the strength of the gauge fixing depending on the

integration time t . We then apply the GFFA to V . The trade off is that we need to calculate the Jacobian of the transformations and propagate forces from $U \rightarrow V$. The following idea will work for whichever flow we use, as long as it takes us towards Landau gauge. The simplest choice would be to flow along the gradient of the gauge-fixing term in Eq. (6). This algorithm is closely related to the [field transformation idea proposed by Luchang](#). A simplified explanation of how this accelerates sampling can be found in Appendix C.

TODO

1. Mention Fourier-accelerated gauge-fixing to complete the algorithm [5]

4.1 The idea

Start from the usual PI with Wilson action

$$\langle O \rangle = \int D[U] O(U) e^{-S_W[U]}. \quad (17)$$

We want to find the field transformation

$$U = \mathcal{F}(V), \quad (18)$$

where the gauge degrees of freedom are suppressed in V . The result of this transformation is a determinant

$$D[U] = D[V] \det F[V]. \quad (19)$$

In this case, the flow $\mathcal{F}$ should actually “coarsen” the gauge degrees of freedom, that is to say it takes us further from Landau gauge. Applying the inverse flow $V = \mathcal{F}^{-1}(U)$ then takes us from an ungauged-fixed configuration U to the Landau gauge-fixed configuration V . Which direction is the forward/reverse flow is arbitrary, but the reverse flow requires fixed-point iterations and so we want to use the reverse flow as infrequently as possible. This scheme only requires a single reverse flow at the beginning of the first trajectory.

The PI becomes

$$\begin{aligned} \langle O \rangle &= \int D[V] [\det \mathcal{F}(V)] O(U(V)) e^{-S_W[U(V)]} \\ &= \int D[V] O(U(V)) e^{-S_W[U(V)] + \log \det \mathcal{F}(V)}. \end{aligned} \quad (20)$$

The term $\log \det F(V)$ mimics the gauge fixing term we would add to the action to fix to Landau gauge, but here we avoid the integral over the gauge transformations. Since the argument of the exponent looks like a gauge-fixed action, we propose to use the GFFA to generate field configurations V . In the usual FTHMC, the physical target configurations U can be obtained by applying the flow to V . For our case, the flow just generates gauge transformations, and therefore we can just measure the gauge-fixed links V directly. This scheme requires that the field transformation has a unique inverse. Martin Lüscher proved that such an inverse transformation exists [6].

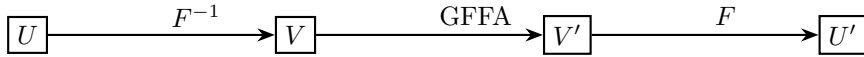


Figure 1: Proposed method for exact GFFA.

4.2 HMC

After making the field transformation, we introduce momenta conjugate to the gauge-fixed variables V . The Hamiltonian is

$$H(\pi, V) = \frac{1}{2}(\pi, K(k^2)\pi) + S_W(\mathcal{F}(V)) - \log \det \mathcal{F}(V). \quad (21)$$

The kinetic kernel $K(k^2)$ is a function of the Fourier modes of the momenta field, equivalent to the kinetic term described in Ref. [1]. We need to evaluate the forces from the gauge fields. Both of the gauge field dependent terms are complicated functions of the gauge-fixed links V , so the method described by Lüscher is to evaluate the forces with respect to the ungauged-fixed links U , then to use the iterative field transformation to map these forces to the transformed links V [6]. In our case, however, the field transformation steps are gauge transformations. This means

$$S(U) = S(\mathcal{F}(V)) = S(V). \quad (22)$$

Therefore, we can actually just evaluate the Wilson action and its derivatives in the usual way with the gauge fixed links V . This will lead to the same approximate conservation of the function $S(U) = S(V)$ as if the forces were propagated, so we'll still get the same acceptance.

It does appear that we still need the force propagation for derivatives of the $\ln \det \mathcal{F}$ terms. I am hoping to take advantage of the gauge-transformation properties to mitigate this overhead.

4.3 Choice of flow

Here I describe the flow which will take us to Landau gauge. To perform the inverse transformation as infrequently as possible, we actually want to use this flow in reverse so that the inverse flow takes us to Landau gauge, and the forward flow may be used to return to the ungauged fixed links when required.

In Ref. [4], they mention a gauge-fixing flow

$$\dot{A}_t = b[A_t], \quad (23)$$

$$b_{x,\mu}^a[A] = \partial_\mu(\partial \cdot A(x))^a + f_{bc}^a A^b(x)(\partial \cdot A(x))^c = \left[D_\mu(A)(\partial \cdot A) \right]^a. \quad (24)$$

This flow was found by Zwanziger [7, 8]. It is the generator of local gauge transformations and is therefore tangent to the gauge orbit, so that gauge-invariant quantities are unchanged along its trajectories. Furthermore, the stable equilibrium points of the flow b are the local minima of Eq. (5). A difficulty is that this force is non-conservative, and so cannot be represented as a gradient flow, but I think this is a minor issue which can be overcome without excess difficulty. The flow is still reversible and so it will work for our purposes. That being said, any gradient descent scheme which gets us to Landau gauge may be appropriate in place of this flow.

The following reproduces the flow in the continuum limit:

$$\Omega(U, x) = \partial_\mu A_\mu(x) = \frac{1}{2} \sum_\mu \left[\left(U(x, \mu) - U(x - \hat{\mu}, \mu)^\dagger \right) - \left(U(x, \mu) - U(x - \hat{\mu}, \mu)^\dagger \right)^\dagger \right]_{\text{traceless}}, \quad (25)$$

The right side of this equation is the general for the projection of any matrix M into $\mathfrak{su}(3)$:

$$\mathcal{P}(M) = \frac{1}{2}(M - M^\dagger) - \frac{1}{6}\text{tr}(M - M^\dagger) = - \sum_a \text{Re} \text{tr}(T^a M) T^a = - \sum_a \text{tr}(T^a (M - M^\dagger)) T^a. \quad (26)$$

Via the right-most equality we can define the components of $\Omega(x)$ in the Lie algebra: $\Omega^a(x) = \text{Retr} T^a(M - M^\dagger)$, where $M = \sum_\mu (U(x, \mu) - U^\dagger(x - \mu, \mu))$.

The equation of motion for the link variables is

$$\dot{U}(x, \mu) = D_\mu \Omega(U, x) = U(x, \mu) \Omega(x + \mu) - \Omega(x) U(x, \mu). \quad (27)$$

In the weak-coupling limit this reduces to the target flow. It is clear that the solution to this DEQ takes the form of a gauge transformation

$$U_t(x, \mu) = e^{-t\Omega(U, x)} U_0(x, \mu) e^{+t\Omega(U, x+\mu)}. \quad (28)$$

This is the same gauge transformation used in Ref. [5] to fix configurations to Landau gauge. I'll drop the U from the argument of Ω going forward. Despite the fact that this flow is a bit more complicated than those described by Lüscher, since $\dot{U} \in \mathfrak{su}(3)$ and it is a differentiable function of t and U , all the uniqueness and existence theorems apply so that this flow can be reversed (See chpt. 35 of [9]).

Applying the flow at a single site x we have the DEQ

$$\begin{aligned} U_t(x, \mu) &= e^{-t\Omega(x)} U(x, \mu), \\ U_t(x - \hat{\mu}, \mu) &= U(x - \hat{\mu}, \mu) e^{t\Omega(x)}. \end{aligned} \quad (29)$$

Obviously the composition of many reversible maps of this type is also reversible.

4.4 Invertability of the flow

We want to prove that we can invert the flow in Eq. (29) with a fixed-point iteration that converges to a unique point. The proof of this follows closely that of the Wilson flow in the appendices of Ref. [6]. We consider a gauge transformation at a single site as described in the previous section. Denote by U' the gauge transformed links.

The generator of the flow is

$$f(X) = \Omega(U, x), \quad (30)$$

evaluated at

$$\begin{aligned} U(x, \mu = 0, \dots, 3) &= e^{+\epsilon X} U'(x, \mu = 0, \dots, 3), \\ U(x - \hat{\mu}, \mu = 0, \dots, 3) &= U'(x - \hat{\mu}, \mu = 0, \dots, 3) e^{-\epsilon X}. \end{aligned} \quad (31)$$

Consider that Ω is a projection into the Lie algebra $\mathfrak{su}(3)$:

$$\Omega(U, x) = \mathcal{P} \left(\sum_\mu \left[U(x, \mu) - U(x - \hat{\mu}, \mu)^\dagger \right] \right). \quad (32)$$

Since $X, \Omega \in \mathfrak{su}(3)$, $f : \mathfrak{su}(3) \rightarrow \mathfrak{su}(3)$.

I will now list some formulae to be used in the proof. In the following $\|M\|$ is the matrix norm satisfying

$$\|A + B\| \leq \|A\| + \|B\|. \quad (33)$$

For $X, Y \in \mathfrak{su}(3)$

$$\|e^X - e^Y\| \leq \|X - Y\|. \quad (34)$$

For any 3×3 complex matrix

$$\|\mathcal{P}(M)\| \leq \frac{4}{3} \|M\|. \quad (35)$$

For $A, B \in SU(3)$

$$\left\| \mathcal{P} \left(A \left[e^X - e^Y \right] B \right) \right\| \leq \frac{4}{3} \|X - Y\|. \quad (36)$$

We want to prove

$$\|f(X) - f(Y)\| \leq k \|X - Y\|, \quad k < 1. \quad (37)$$

If the flow satisfies this, it is a strict contraction. Strict contractions in complete metric spaces have a unique fixed point (see for example Chapter II.C of Ref. [10]). The fixed point X_* is reached via the iteration $X_{n+1} = f(X_n)$, which will converge at an exponential rate. Thus, the matrices

$$\begin{aligned} U(x, \mu = 0, \dots, 3) &= e^{+\epsilon X_*} U'(x, \mu = 0, \dots, 3), \\ U(x - \hat{\mu}, \mu = 0, \dots, 3) &= U'(x - \hat{\mu}, \mu = 0, \dots, 3) e^{-\epsilon X_*}, \end{aligned} \quad (38)$$

are the unique inverses we seek.

The proof is as follows

$$\begin{aligned} \|f(X) - f(Y)\| &= \left\| \mathcal{P} \left\{ \sum_{\mu} \left(e^{\epsilon X} U(x, \mu) - e^{\epsilon X} U^\dagger(x - \hat{\mu}, \mu) \right) \right\} - \mathcal{P} \left\{ \sum_{\mu} \left(e^{\epsilon Y} U(x, \mu) - e^{\epsilon Y} U^\dagger(x - \hat{\mu}, \mu) \right) \right\} \right\| \\ &= \left\| \mathcal{P} \left\{ \sum_{\mu} \left[\left(e^{\epsilon X} - e^{\epsilon Y} \right) U(x, \mu) - \left(e^{\epsilon X} - e^{\epsilon Y} \right) U^\dagger(x - \hat{\mu}, \mu) \right] \right\} \right\| \end{aligned} \quad (39)$$

We use Eq. (33) for the sums over μ and to split up the two terms inside the projection $\mathcal{P}$.

$$\begin{aligned} \|f(X) - f(Y)\| &\leq \sum_{\mu} \left[\left\| \mathcal{P} \left\{ \left(e^{\epsilon X} - e^{\epsilon Y} \right) U(x, \mu) \right\} \right\| + \left\| \mathcal{P} \left\{ \left(e^{\epsilon X} - e^{\epsilon Y} \right) U^\dagger(x - \hat{\mu}, \mu) \right\} \right\| \right] \\ &\leq \sum_{\mu} \left[\frac{4}{3} |\epsilon| \|X - Y\| + \frac{4}{3} |\epsilon| \|X - Y\| \right] \\ &= \frac{32}{3} |\epsilon| \|X - Y\|. \end{aligned} \quad (40)$$

To reach the second line we used Eq. (36), and to reach the third we used Eq. (35). **(Need to check the coefficient to be sure I didn't miss any factors)**

Therefore, we find that the generator of the flow is a strict contraction for discrete time steps $|\epsilon| < 3/32$.

4.5 Jacobian matrix

To get an idea of the Jacobian of a gauge transformation at site x $e^{\epsilon X(x)}$, we consider separately the left- and right-applications $e^{\epsilon X} U(x, \mu)$ and $U(x - \hat{\mu}, \mu) e^{-\epsilon X}$ separately. For the first, we use a basis of right-invariant differential operators to calculate the Jacobian. For the second, we use a left-invariant basis. See Appendix F for my L/R invariant derivative operator convention. The total Jacobian is more complicated than the product of these, but this calculation demonstrates that the leading part of the determinant provides the soft-gauge fixing we're interested in.

4.5.1 Left application

Here we calculate the Jacobian of the single step in Eq. (??). We use the methods described in detail in Ref. [6]. For the left-application, the Jacobian matrix is

$$\left[F(U)\right](x, \mu; y, \nu)^{ab} = -2 \text{Tr} \left\{ \partial_{x, \mu}^b U^L(y, \nu) \left[U^L(y, \nu)\right]^{-1} T^a \right\}, \quad (41)$$

Corresponding to the basis of right-invariant differential operators. Our updates will be a series of single-link field transformations

$$\begin{aligned} \left[\mathcal{E}(U)_{x, \mu}\right](y, \nu)^{ab} &= e^{-t\Omega(x)} U(y, \nu) \quad \text{if } (x, \mu) = (y, \nu) \\ &= U(y, \nu) \quad \text{otherwise.} \end{aligned} \quad (42)$$

This means that the total Jacobian is the ordered product of the single-link Jacobians

$$\left[\mathcal{E}(U)_{x, \mu, *}\right](y, \nu; z, \rho)^{ac} = -2 \text{Tr} \left\{ \partial_{z, \rho}^c \left[\mathcal{E}(U)_{x, \mu}\right](y, \nu) \left[\mathcal{E}(U)_{x, \mu}\right](y, \nu)^{-1} T^a \right\}. \quad (43)$$

We can express this via the derivative of $\Omega(x)$ with respect to the link variables

$$\left[O_*(U)\right](y; z, \rho)^{bc} = \partial_{z, \rho}^c \Omega^b(y). \quad (44)$$

The explicit expression is

$$\begin{aligned} \left[\mathcal{E}(U)_{x, *}\right](y, \nu; z, \rho)^{ac} &= \delta^{ac} \delta_{yz} \delta_{\nu\rho} \\ &+ \delta_{xy} \delta_{\mu\nu} \left\{ \left(e^{\text{Ad } X} - 1\right)^{ac} \delta_{yz} \delta_{\nu\rho} + \tau J(-X)^{ab} \left[O_*(U)\right](y; z, \rho)^{bc} \right\}_{X=\tau\Omega(x)}, \end{aligned} \quad (45)$$

which is clearly non-vanishing for our choice of field transformation.

We can also compute the determinant of each individual Euler step as in Appendix C of Ref. [6]. A gauge transformation on a single site changes all the links to which it is connected. Here we just want to point out that the leading contribution from the determinant is the soft gauge fixing term, so we consider just a single link update. In this case, only $(x, \mu) = (y, \nu)$ row of the Jacobian matrix is non-trivial, with identity in all other diagonal entries and the remaining off diagonal terms vanish. We consider in the following the determinant of a single update step, and show that it does not vanish. As we will see in the following sections this calculation gives the right leading ϵ behavior. The only relevant element of Eq. (45) is (I changed the sign of t above, need to be careful to update the following to reflect this.)

$$A^{ac} = \left\{ \left(e^{\text{Ad } X}\right)^{ac} + \tau J(-X)^{ab} \left[O_*(U)\right](x; x, \mu)^{bc} \right\}_{X=\tau\Omega(x)}. \quad (46)$$

We can expand to linear order in the time-step to find the leading effect

$$\begin{aligned} A^{ac} &= \left\{ \delta^{ac} + \tau f^{ach} \Omega^h(x) + \tau \partial_{x, \mu}^c \Omega^b(x) \right\} \\ &= \left\{ \delta^{ac} + \tau f^{ach} \Omega^h(x) - \tau \delta^{ab} \sum_{\nu} \partial_{x, \mu}^c \text{tr} \left[T^b \left(\left\{ U(x, \nu) - U^\dagger(x - \hat{\nu}, \nu) \right\} - \left\{ U(x, \nu) - U^\dagger(x - \hat{\nu}, \nu) \right\}^\dagger \right) \right] \right\} \\ &= \left\{ \delta^{ac} + \tau f^{ach} \Omega^h(x) - \tau \text{tr} \left[\left(T^a T^c U(x, \mu) + T^c T^a U(x, \mu)^\dagger \right) \right] \right\}. \end{aligned} \quad (47)$$

Consider

$$\det A = \frac{1}{n!} \epsilon_{i_1 i_2 \dots i_n} \epsilon_{j_1 j_2 \dots j_n} A_{i_1 j_1} A_{i_2 j_2} \dots A_{i_n j_n}. \quad (48)$$

Because we're just looking for leading order in τ , we just need to take a single of the A_{lk} to be proportional to τ , and the rest are δ_{lk} . Using

$$\epsilon_{a i_2 \dots i_n} \epsilon_{b i_2 \dots i_n} = (n-1)! \cdot \delta_{ab}, \quad (49)$$

We find

$$\begin{aligned} \det A &= 1 + \frac{1}{8} \delta_{ac} \left\{ \tau f^{ach} \Omega^h(x) - \tau \text{tr} \left[T^a T^c U(x, \mu) + T^c T^a U(x, \mu)^\dagger \right] \right\} + O(\tau^2) \\ &= 1 - \frac{\tau}{8} \left(2 \frac{-16}{3} \right) \text{Retr} U(x, \mu) + O(\tau^2) \\ &= 1 + \frac{4}{3} \tau \text{Retr} U(x, \mu) + O(\tau^2) \end{aligned} \quad (50)$$

which is exactly what we want to suppress fluctuations away from Landau gauge since $\log \det A \propto \text{Retr} U$. In the final line we used

$$\sum_a (T^a T^a)_{ij} = -\frac{16}{3} \delta_{ij}. \quad (51)$$

A more detailed calculation which proves that this determinant never vanishes can be found in the appendices of Ref. [6].

4.5.2 Right application

We also need the Jacobian of the other right step. For this one, we actually need to consider left-invariant differential operators and the corresponding basis of left-invariant differential forms. The Jacobian in this case is:

$$\left[F(U) \right] (x, \mu; y, \nu)^{ab} = -2 \text{Tr} \left\{ \left[U^R(y, \nu) \right]^{-1} \partial_{x, \mu}^b U^R(y, \nu) T^a \right\}. \quad (52)$$

Using the Euler integrator would result in the same Jacobian as in Eq. (45), except with the adjoint operators $\text{Ad} X \rightarrow \text{Ad}(-X)$. We actually already have a minus sign in the right updates because of the finite difference, so the Jacobian is only slightly changed

$$\begin{aligned} \left[\mathcal{E}(U)_{x,*} \right] (y, \nu; z, \rho)^{ac} &= \delta^{ac} \delta_{yz} \delta_{\nu\rho} \\ &+ \delta_{xy} \delta_{\mu\nu} \left\{ \left(e^{\text{Ad } X} - 1 \right)^{ac} \delta_{yz} \delta_{\nu\rho} - \tau J(X)^{ab} \left[O_*(U) \right] (y; z, \rho)^{bc} \right\}_{X=\tau\Omega(x+\hat{\mu})}, \end{aligned} \quad (53)$$

Another difference is

$$\left[O_*(U) \right] (y; z, \rho)^{bc} = \partial_{z, \rho}^c \Omega^b(y). \quad (54)$$

In this case we evaluate

$$\begin{aligned} \partial_{x, \mu}^c \Omega^b(x + \mu) &= \sum_{\nu} \partial_{x, \mu}^c \text{tr} \left[T^b \left(\left\{ U(x + \mu, \nu) - U^\dagger(x + \hat{\mu} - \hat{\nu}, \nu) \right\} - \left\{ U(x + \mu, \nu) - U^\dagger(x + \hat{\mu} - \hat{\nu}, \nu) \right\}^\dagger \right) \right] \\ &= \sum_{\nu} \text{tr} \left[T^b \left(- \left\{ U(x + \hat{\mu} - \hat{\nu}, \nu) T^c \right\}^\dagger + U(x + \hat{\mu} - \hat{\nu}, \nu) T^c \right) \delta_{\nu\mu} \right] \\ &= \text{tr} \left[T^b T^c U(x, \mu)^\dagger + T^c T^b U(x, \mu) \right] \end{aligned} \quad (55)$$

Combining this with the $-\tau J$ piece and calculating the determinant as before, while keeping the leading order in τ we find

$$\det A = 1 + \frac{4}{3}\tau \text{Retr} U(x, \mu) + O(\tau^2). \quad (56)$$

Which again gives us the suppression of gauge fluctuations.

4.6 Full Jacobian matrix

(Now using correct signs on flow time t) The Haar measure can be defined equally well with right- or left-invariant basis of one-forms. We can take advantage of this to simplify the Jacobian calculations. Consider the discrete updates

$$\begin{aligned} U(x, \mu) &\rightarrow U'(x, \mu) = e^{-\epsilon\Omega(x)} U(x, \mu), \\ U(x - \hat{\mu}, \mu) &\rightarrow U'(x - \hat{\mu}, \mu) = U(x - \hat{\mu}, \mu) e^{\epsilon\Omega(x)}. \end{aligned} \quad (57)$$

We choose the one-form basis associated with links at (x, μ) which are updates from the left to be right-invariant and those which are at the locations $(x - \hat{\mu}, \mu)$ and updates from the right to be left-invariant. Thus, denoting by $\theta_{x, \mu} = \theta_{x, \mu}^a T^a$ the one-forms at these sites, we write

$$\begin{aligned} \theta_{y, \nu} &= dU'(y, \nu) U'^{-1}(y, \nu) \delta_{(y, \nu), RI} + U'^{-1}(y, \nu) dU'(y, \nu) \delta_{(y, \nu), LI} \\ &= d\left(e^{X_L} U(y, \nu)\right) \left(e^{X_L} U(y, \nu)\right)^{-1} \delta_{(y, \nu), RI} + \left(U(y, \nu) e^{X_R}\right)^{-1} d\left(U(y, \nu) e^{X_R}\right) \delta_{(y, \nu), LI} \end{aligned} \quad (58)$$

Write the exterior derivative d in terms of the right- or left-invariant basis corresponding to the basis U

$$d = \alpha_{z, \rho}^b \partial_{z, \rho}^{L, b} \delta_{(z, \rho), LI} + \beta_{z, \rho}^b \partial_{z, \rho}^{R, b} \delta_{(z, \rho), RI}, \quad (59)$$

where

$$\begin{aligned} \alpha_{x, \mu} &= U^\dagger(x, \mu) dU(x, \mu) \\ \beta_{x, \mu} &= dU(x, \mu) U^\dagger(x, \mu) \end{aligned} \quad (60)$$

We have then

$$\begin{aligned} \theta_{y, \nu} &= \left[\alpha_{z, \rho}^b \partial_{z, \rho}^{L, b} \delta_{(z, \rho), LI} + \beta_{z, \rho}^b \partial_{z, \rho}^{R, b} \delta_{(z, \rho), RI} \right] \left(e^{X_L} U(y, \nu) \right) \left(U^\dagger(y, \nu) e^{-X_L} \right) \delta_{(y, \nu), RI} \\ &\quad + \left(e^{-X_R} U^\dagger(y, \nu) \right) \left[\alpha_{z, \rho}^b \partial_{z, \rho}^{L, b} \delta_{(z, \rho), LI} + \beta_{z, \rho}^b \partial_{z, \rho}^{R, b} \delta_{(z, \rho), RI} \right] \left(U(y, \nu) e^{X_R} \right) \delta_{(y, \nu), LI} \end{aligned} \quad (61)$$

Both ∂^L and ∂^R act on the e^x terms, but for the $U(x, \mu)$ terms we have, for example

$$\begin{aligned} &\beta_{z, \rho}^b \partial_{z, \rho}^{R, b} \delta_{(z, \rho), RI} U(y, \nu) \delta_{(y, \nu), LI} \\ &= \beta_{z, \rho}^b \delta_{(z, \rho), RI} \delta_{(y, \nu), LI} \partial_{z, \rho}^{R, b} U(y, \nu) \\ &= \beta_{z, \rho}^b \delta_{(z, \rho), RI} \delta_{(y, \nu), LI} T^b U(y, \nu) \delta_{z, y} \delta_{\rho, \nu} \\ &= \delta_{(y, \nu), RI} \delta_{(y, \nu), LI} \beta_{y, \nu}^b T^b U(y, \nu) \\ &\propto \delta_{LI, RI} = 0. \end{aligned} \quad (62)$$

So we can drop all terms like this. This leaves

$$\begin{aligned}
\boldsymbol{\theta}_{\mathbf{y},\boldsymbol{\nu}} = & \delta_{(y,\nu),RI} \left\{ \delta_{(z,\rho),LI} \alpha_{z,\rho}^b \partial_{z,\rho}^{L,b} e^{X_L(y,\nu)} e^{-X_L(y,\nu)} \right. \\
& + \beta_{z,\rho}^b \delta_{(z,\rho),RI} \left[\partial_{z,\rho}^{R,b} e^{X_L(y,\nu)} e^{-X_L(y,\nu)} + e^{X_L(y,\nu)} \partial_{z,\rho}^{R,b} U(y,\nu) U^\dagger(y,\nu) e^{-X_L(y,\nu)} \right] \Big\} \\
& + \delta_{(y,\nu),LI} \left\{ \alpha_{z,\rho}^b \delta_{(z,\rho),LI} \left[e^{-X_R} \partial_{z,\rho}^{L,b} e^{X_R} + e^{-X_R} U^\dagger(y,\nu) \partial_{z,\rho}^{L,b} U(y,\nu) e^{X_R} \right] \right. \\
& + \beta_{z,\rho}^b \delta_{(z,\rho),RI} e^{-X_R} \partial_{z,\rho}^{R,b} e^{X_R} \Big\}
\end{aligned} \tag{63}$$

Consider

$$0 = \frac{d}{dt} \left[e^{-X(t)} e^{X(t)} \right] = e^{-X(t)} \frac{d}{dt} e^{X(t)} + \frac{d}{dt} e^{-X(t)} e^{X(t)}.$$
(64)

We have

$$\partial_{x,\mu}^a e^X e^{-X} = J(-X) \cdot X_*,$$
(65)

implying

$$e^{-X} \partial_{x,\mu}^a e^X = +J(X) \cdot X_*.$$
(66)

useful property of $J(X)$:

$$J^T(X) = J(-X) = e^{\text{adj}(X)} J(X).$$
(67)

We also use

$$e^X dU U^\dagger e^{-X} = e^{\text{Adj}(X)} dU U^\dagger.$$
(68)

Therefore we have

$$\begin{aligned}
\boldsymbol{\theta}_{\mathbf{y},\boldsymbol{\nu}} = & \delta_{(y,\nu),RI} \left\{ \delta_{(z,\rho),LI} \alpha_{z,\rho}^b J^b(-X_L) \cdot X_{L*}^L(z,\rho;y,\nu) \right. \\
& + \beta_{z,\rho}^b \delta_{(z,\rho),RI} \left[J^b(-X_L) \cdot X_{L*}^R(z,\rho;y,\nu) + \left(e^{\text{Adj} X_L(y,\nu)} \right)^{bc} \partial_{z,\rho}^{R,c} U(y,\nu) U^\dagger(y,\nu) \right] \Big\} \\
& + \delta_{(y,\nu),LI} \left\{ \alpha_{z,\rho}^b \delta_{(z,\rho),LI} \left[J^b(X_R) \cdot X_{R*}^L(z,\rho;y,\nu) + \left(e^{\text{Adj}(-X_L(y,\nu))} \right)^{bc} U^\dagger(y,\nu) \partial_{z,\rho}^{L,c} U(y,\nu) \right] \right. \\
& + \beta_{z,\rho}^b \delta_{(z,\rho),RI} J^b(X_R) \cdot X_{R*}^R(z,\rho;y,\nu) \Big\}.
\end{aligned} \tag{69}$$

Here, $X_L = -\epsilon\Omega$, $X_R = \epsilon\Omega$, $X_{B*}^A = \partial^A X_B$ (Hopefully these will can be easily calculated using

auto-differentiation code):

$$\begin{aligned}
X_{R*}^R &= \partial_{z,\rho}^{R,a} X_R(y)^b T^b = \epsilon \partial_{z,\rho}^{R,a} \sum_{\nu} \text{tr} \left[T^b \left(U(y, \nu) - U^\dagger(y - \hat{\nu}, \nu) - \left\{ U^\dagger(y, \nu) - U(y - \hat{\nu}, \nu) \right\} \right) \right] T^b \\
&= \epsilon \sum_{\nu} \text{tr} \left[T^b T^a U(y, \nu) + T^a T^b U^\dagger(y, \nu) \right] T^b \delta_{z,y} \delta_{\rho,\nu} \\
&= 2\epsilon \sum_{\nu} \text{Retr} \left[T^b T^a U(y, \nu) \right] T^b \delta_{z,y} \delta_{\rho,\nu} \\
&= 2\epsilon \sum_{\nu} \mathcal{P} \left\{ T^a U(y, \nu) \right\} \delta_{z,y} \delta_{\rho,\nu} \\
&= 2\epsilon \mathcal{P} \left\{ T^a U(y, \nu) \right\} \delta_{z,y} \delta_{\rho,\nu}
\end{aligned} \tag{70}$$

$$\begin{aligned}
X_{R*}^L &= \partial_{z,\rho}^{L,a} X_R(y)^b T^b = \epsilon \partial_{z,\rho}^{L,a} \sum_{\nu} \text{tr} \left[T^b \left(U(y, \nu) - U^\dagger(y - \hat{\nu}, \nu) - \left\{ U^\dagger(y, \nu) - U(y - \hat{\nu}, \nu) \right\} \right) \right] T^b \\
&= \epsilon \sum_{\nu} \text{tr} \left[-T^b T^a U^\dagger(y - \hat{\nu}, \nu) - T^a T^b U(y - \hat{\nu}, \nu) \right] T^b \delta_{z,y-\hat{\nu}} \delta_{\rho,\nu} \\
&= -2\epsilon \mathcal{P} \left\{ U(y - \hat{\nu}, \nu) T^a \right\} \delta_{z,y-\hat{\nu}} \delta_{\rho,\nu}
\end{aligned} \tag{71}$$

$$\begin{aligned}
X_{L*}^L &= \partial_{z,\rho}^{L,a} X_L(y)^b T^b = -\epsilon \partial_{z,\rho}^{L,a} \sum_{\nu} \text{tr} \left[T^b \left(U(y, \nu) - U^\dagger(y - \hat{\nu}, \nu) - \left\{ U^\dagger(y, \nu) - U(y - \hat{\nu}, \nu) \right\} \right) \right] T^b \\
&= -\epsilon \sum_{\nu} \text{tr} \left[-T^b T^a U^\dagger(y - \hat{\nu}, \nu) - T^a T^b U(y - \hat{\nu}, \nu) \right] T^b \delta_{z,y-\hat{\nu}} \delta_{\rho,\nu} \\
&= 2\epsilon \mathcal{P} \left\{ U(y - \hat{\nu}, \nu) T^a \right\} \delta_{z,y-\hat{\nu}} \delta_{\rho,\nu}
\end{aligned} \tag{72}$$

$$\begin{aligned}
X_{L*}^R &= \partial_{z,\rho}^{R,a} X_L(y)^b T^b = -\epsilon \partial_{z,\rho}^{R,a} \sum_{\nu} \text{tr} \left[T^b \left(U(y, \nu) - U^\dagger(y - \hat{\nu}, \nu) - \left\{ U^\dagger(y, \nu) - U(y - \hat{\nu}, \nu) \right\} \right) \right] T^b \\
&= -\epsilon \sum_{\nu} \text{tr} \left[T^b T^a U(y, \nu) + T^a T^b U^\dagger(y, \nu) \right] T^b \delta_{z,y} \delta_{\rho,\nu} \\
&= -2\epsilon \mathcal{P} \left\{ T^a U(y, \nu) \right\} \delta_{z,y} \delta_{\rho,\nu}
\end{aligned} \tag{73}$$

We can also simplify the one-form basis expression to include explicit factors of T^a

$$\begin{aligned}
\theta_{y,\nu}^d T^d &= \delta_{(y,\nu),RI} \left\{ \delta_{(z,\rho),LI} \alpha_{z,\rho}^b J^{bc} (-X_L) X_{L*}^L(z, \rho; y, \nu)^{cd} \right. \\
&\quad \left. + \beta_{z,\rho}^b \delta_{(z,\rho),RI} \left[J^{bc} (-X_L) X_{L*}^R(z, \rho; y, \nu)^{cd} + \left(e^{\text{Adj} X_L(y,\nu)} \right)^{bd} \delta_{z,y} \delta_{\rho,\nu} \right] \right\} T^d \\
&\quad + \delta_{(y,\nu),LI} \left\{ \alpha_{z,\rho}^b \delta_{(z,\rho),LI} \left[J^{bc} (X_R) X_{R*}^L(z, \rho; y, \nu)^{cd} - \left(e^{\text{Adj} (-X_R(y,\nu))} \right)^{bd} \delta_{y,z} \delta_{\rho,\nu} \right] \right. \\
&\quad \left. + \beta_{z,\rho}^b \delta_{(z,\rho),RI} J^{bc} (X_R) \cdot X_{R*}^R(z, \rho; y, \nu)^{cd} \right\} T^d.
\end{aligned} \tag{74}$$

Now we can collect one-forms for U links α and β into a collective basis

$$\phi_{x,\mu} = \phi_{x,\mu}^d T^d = \alpha_{x,\mu} \delta_{(x,\mu),LI} + \beta_{x,\mu} \delta_{(x,\mu),RI}. \quad (75)$$

This lets us write the Jacobian $\mathcal{F}(x)$ for the change of variables $U/\phi \rightarrow U'/\theta$ at site x as

$$\begin{aligned} \theta_{y,\nu}^d &= \sum [\mathcal{F}(x)](y, \nu; z, \rho)^{ad} \phi_{z,\rho}^a \\ [\mathcal{F}(x)](y, \nu; z, \rho)^{ad} &= \delta_{(y,\nu),RI} \left\{ \delta_{(z,\rho),LI} J^{ac}(-X_L) X_{L*}^L(z, \rho; y, \nu)^{cd} \right. \\ &\quad \left. + \delta_{(z,\rho),RI} \left[J^{ac}(-X_L) X_{L*}^R(z, \rho; y, \nu)^{cd} + \left(e^{\text{Adj} X_L(y,\nu)} \right)^{ad} \delta_{z,y} \delta_{\nu,\rho} \right] \right\} \\ &\quad + \delta_{(y,\nu),LI} \left\{ \delta_{(z,\rho),LI} \left[J^{ac}(X_R) X_{R*}^L(z, \rho; y, \nu)^{cd} - \left(e^{\text{Adj}(-X_R(y,\nu))} \right)^{ad} \delta_{y,z} \delta_{\rho,\nu} \right] \right. \\ &\quad \left. + \delta_{(z,\rho),RI} J^{ac}(X_R) \cdot X_{R*}^R(z, \rho; y, \nu)^{cd} \right\}. \end{aligned} \quad (77)$$

With $(x, \mu) = RI$, $(x - \hat{\mu}, \mu) = LI$ for all $1 \leq \mu \leq 4$. The minus sign on the e^{Adj} term in the third line looks a bit strange, but the same minus sign occurs for the left-invariant basis form in the U' basis so its actually fine.

We can take advantage of the fact that $X_R = -X_L$ to factor out $J(X_R)$

$$\begin{aligned} [\mathcal{F}(x)](y, \nu; z, \rho)^{ad} &= J^{ac}(X_R) \delta_{(y,\nu),RI} \left\{ \delta_{(z,\rho),LI} X_{L*}^L(z, \rho; y, \nu)^{cd} \right. \\ &\quad \left. + \delta_{(z,\rho),RI} \left[X_{L*}^R(z, \rho; y, \nu)^{cd} + \left[J(X_R)^{-1} \right]^{cf} \left(e^{\text{Adj} X_L(y,\nu)} \right)^{fd} \delta_{z,y} \delta_{\nu,\rho} \right] \right\} \\ &\quad + J^{ac}(X_R) \delta_{(y,\nu),LI} \left\{ \delta_{(z,\rho),LI} \left[X_{R*}^L(z, \rho; y, \nu)^{cd} - \left[J(X_R)^{-1} \right]^{cd} \left(e^{\text{Adj}(-X_R(y,\nu))} \right)^{fd} \delta_{y,z} \delta_{\rho,\nu} \right] \right. \\ &\quad \left. + \delta_{(z,\rho),RI} X_{R*}^R(z, \rho; y, \nu)^{cd} \right\}. \end{aligned} \quad (78)$$

We can simplify further

$$\begin{aligned} \left[J(X_R)^{-1} \right] \left(e^{\text{Adj}(-X_R(y,\nu))} \right) &= \left[\left(e^{\text{Adj}(+X_R(y,\nu))} \right) J(X_R) \right]^{-1} \\ &= \left[J(-X_R) \right]^{-1}. \end{aligned} \quad (79)$$

Therefore

$$\begin{aligned}
\left[\mathcal{F}(x)\right](y, \nu; z, \rho)^{ad} = & J^{ac}(X_R) \delta_{(y, \nu), RI} \left\{ \delta_{(z, \rho), LI} X_{L*}^L(z, \rho; y, \nu)^{cd} \right. \\
& \left. + \delta_{(z, \rho), RI} \left[X_{L*}^R(z, \rho; y, \nu)^{cd} + \left[J(-X_R)^{-1} \right]^{cd} \delta_{z, y} \delta_{\nu, \rho} \right] \right\} \\
& + J^{ac}(X_R) \delta_{(y, \nu), LI} \left\{ \delta_{(z, \rho), LI} \left[X_{R*}^L(z, \rho; y, \nu)^{cd} - \left[J(-X_R)^{-1} \right]^{cd} \delta_{y, z} \delta_{\rho, \nu} \right] \right. \\
& \left. + \delta_{(z, \rho), RI} X_{R*}^R(z, \rho; y, \nu)^{cd} \right\}.
\end{aligned} \tag{80}$$

This expression can therefore be calculated from $J(X_R)$ and $X_{R*} = \text{Retr}(T^a U)$, with X_{L*} reached by cshifts on X_{R*} .

4.7 Jacobian determinant

4.7.1 Leading effect

The full form of this determinant is a bit tricky, but we can extract its small ϵ behavior

$$\det \left(1 + \epsilon A \right) \underset{\epsilon \rightarrow 0}{=} 1 + \epsilon \text{tr} A + O(\epsilon^2). \tag{81}$$

We should just get the same terms as calculated in Sections 4.5.1 and 4.5.2.

4.7.2 Determinant calculation

We can split the updates into those performed at even and odd sites. The updates at all even sites are decoupled from each other and are well suited to parallel computation. The factorization of the Jacobian means that for one even step we have

$$\det \mathcal{F} = \prod_{x \in \text{even}} \det \mathcal{F}(x). \tag{82}$$

When we eventually take the logarithm of this we'll have the sum of the individual pieces

$$\ln \det \mathcal{F} = \sum_{x \in \text{even}} \ln \det \mathcal{F}(x). \tag{83}$$

4.8 Force from Jacobian

We need to calculate

$$d \ln \det \left[\mathcal{F}(x) \right] (y, \nu; z, \rho)^{ad}. \tag{84}$$

Consider Jacobi's formula

$$d \det A = \text{tr} \text{Adj}(A) dA = \det A \text{tr} \left(A^{-1} dA \right). \tag{85}$$

For us

$$d \ln \det A = \text{tr} \left(A^{-1} dA \right). \tag{86}$$

A Gribov ambiguity

The Gribov ambiguity is the statement that there are multiple points in a gauge orbit where the Lorentz gauge-fixing condition is satisfied: $\partial \cdot A_2 = \partial \cdot A_1$. Here,

$$A_\mu^{(2)} = S^\dagger A_\mu^{(1)} S + S^\dagger \partial_\mu S. \quad (87)$$

Taking the divergence and simplifying, one finds that the gauge transformation which generates the ambiguity satisfies

$$\partial_\mu [S^\dagger [D_\mu(A^{(1)}), S]] = 0. \quad (88)$$

If one considers small GTFs $S = 1 + \omega$, they find

$$\partial_\mu [D_\mu(A^{(1)}), \omega] = 0. \quad (89)$$

This is the FP operator, and we find that it is the (near-)zero modes of this operator which cause the ambiguities. The argument then becomes rather complicated, but the result is that the gluon low-modes actually contribute the Gribov ambiguities, while the high-modes are confined to a region in field space without copies.

B Zwanziger-Parrinello-Lasinio gauge-fixing term

Here I show that the minima of the gauge-fixing term

$$X[A^g] = \frac{\beta M^2}{2} (A, A), \quad (90)$$

satisfies Landau gauge.

$$\begin{aligned} X[A^g] &= \int \text{Tr} \{ A^g \wedge * A^g \} \\ &= \int \text{Tr} \{ (g^\dagger A g + g^\dagger dg) \wedge * (g^\dagger A g + g^\dagger dg) \} \\ &= \int \text{Tr} \{ A \wedge * A + g^\dagger A \wedge * dg + dg \wedge g^\dagger * A + g^\dagger dg \wedge * g^\dagger dg \} \\ &= \int \text{Tr} \{ A \wedge * A + g^\dagger A \wedge * dg + dg \wedge g^\dagger * A + \theta \wedge * \theta \}. \end{aligned} \quad (91)$$

In the final line I've used the fact that $g^\dagger dg$ defines the left-invariant Maurer-Cartan form associated with the gauge group. I have to be careful, however, because the combinations $g^\dagger dg$ actually define the left-invariant form, which are equivalent under conjugation by g . Now we can consider a local parameterization of the gauge group $g = e^\omega$. We just focus on the relevant term containing the gauge field and ω cross term:

$$\begin{aligned} &\int \text{Tr} \{ g^\dagger A \wedge * dg + dg \wedge g^\dagger * A \} \\ &= \int \text{Tr} \{ (1 - \omega) A \wedge * d\omega + d\omega \wedge (1 - \omega) * A \}. \end{aligned} \quad (92)$$

Now we vary X with respect to the gauge parameter ω and focus on the terms linear in ω to find

$$0 = \int \text{Tr}\{A \wedge *d\omega\} = \int \text{Tr}\{\delta A \wedge *\omega\} = \int d^d x \text{Tr}\{\partial A \omega\}. \quad (93)$$

For this to be satisfied we find $\partial A = 0$, which is the Landau gauge condition.

In Ref. [11], Fachin derives the Feynman rules introduced by the soft gauge-fixing $\log I[A]$ term. He finds that the resulting contributions are non-local. He also finds that the radiative corrections don't contribute to the longitudinal polarization, so that this method of gauge fixing doesn't introduce non-renormalizable terms. He finds that the expectation values of gauge-invariant observables don't depend on the soft gauge-fixing parameter M . In the limit $M^2 \rightarrow \infty$, the Feynman rules reduce to the usual "hard" Landau gauge-fixed propagator.

C Usual FTHMC algorithm

Today, 01/28/25, Norman and I had a long discussion of how the FTHMC relates to the other methods of Fourier acceleration, such as the GFFA and RMHMC. The following is a description of how it works.

Consider trying to accelerate the harmonic oscillator portion of YM theory, which we model here as an unconstrained classical SHO:

$$V(q) = \frac{1}{2} \sum_i \omega_i^2 q_i^2. \quad (94)$$

With the usual HMC, we'd introduce conjugate momenta p and choose the Hamiltonian

$$H(p, q) = \sum_i \frac{p_i^2}{2m_i} + \frac{1}{2} \sum_i \omega_i^2 q_i^2, \quad (95)$$

and use the equations of motion

$$\dot{q}_i = \frac{p_i}{m_i}, \quad \dot{p}_i = -\omega_i^2 q_i. \quad (96)$$

Rescaling $q_i \rightarrow \omega_i^2 q_i$ we find

$$\dot{q}_i = \frac{\omega_i^2 p_i}{m_i}, \quad \dot{p}_i = -q_i. \quad (97)$$

We find that the different modes q_i move with characteristic velocity ω_i^2/m_i . This means that there is a large mismatch between the high- and low-frequency modes in terms of rate of evolution in MD time units. As we approach continuum limit there is an ever larger disparity between $\omega_{\max}$ and $\omega_{\min}$ and this is critical slowing down (CSD). If we can choose $m_i = \omega_i^2$, we'll completely eliminate CSD. This is the goal of the GFFA and RMHMC.

A rather different method to do this efficiently was [proposed by Luchang](#). Suppose that we instead find a field transformation which smooths out the q_i . This can be used to change the characteristic frequency of the SHO. In this model we use the smoothing operation

$$\dot{q}_i(t) = -\partial_i V(q) = -\omega_i^2 q_i(t), \quad (98)$$

where t is a "flow time", not to be confused with the MD time unit. For illustrative purposes, I'll just use a single Euler step, not focusing too much on the reversibility aspect. Consider just a single discrete flow step so that we can write the flowed variables as

$$q_i^f(t) = (1 - t\omega_i^2)q_i. \quad (99)$$

This is just a change of variables and thus doesn't effect the potential:

$$V(q(q^f)) = \frac{1}{2}(\omega_i^f)^2(q_i^f)^2 = \frac{1}{2} \sum_i \left[\frac{\omega_i}{(1 - t\omega_i^2)} \right]^2 \left[(1 - t\omega_i^2)q_i \right]^2. \quad (100)$$

However, if we introduce momenta conjugate to the flowed variables q^f we find

$$H(p^f, q^f) = \frac{1}{2} \sum_i (p_i^f)^2 + \frac{1}{2} \sum_i \left[\frac{\omega_i}{(1 - t\omega_i^2)} \right]^2 (q_i^f)^2. \quad (101)$$

Thus, if we do an inverse flow along $-t$, we actually decrease the frequencies ω_i^f . Furthermore, the larger frequencies decrease more quickly. Writing the EoM with the rescaling we find

$$\dot{q}_i^f = \left[\frac{\omega_i}{(1 - t\omega_i^2)} \right]^2 p_i^f, \quad \dot{p}_i^f = -q_i^f. \quad (102)$$

Thus, flowing for acts to bring the effective velocities of the coordinates q_i^f more inline with each other, reducing CSD.

D Properties of gauge orbits

The following is from Zwanziger *Stochastic quantization of gauge theories* in Ref. [8]. Considering the usual gauge transformation of connection A_μ^a in continuum. Define the Hilbert norm

$$\|A\|^2 = \int d^n x \sum_\mu \text{tr}(A_\mu(x))^2. \quad (103)$$

We find then that the Hilbert distance between two connections A_2 and A_1 is gauge-invariant

$$\|A_2^g - A_1^g\| = \|A_2 - A_1\|. \quad (104)$$

in progress

E Gauge fixing integral as 5D pure YM theory in maximal tree gauge

It is interesting to note that the gauge transformation matrix (GTM) $g(x)$ enters eight terms in the sum in S_{GF} . It is tempting to fit these group variables into a 5D lattice pure YM theory, where the two GTMs $g(x)$ and $g(x + \mu)$ are dynamic links and the variables $U(x, \mu)$ are fixed staples.

Consider trying to Fourier accelerate 2D pure gauge theory. I am still trying to work out the geometry to fit the GTMs onto a 3D lattice. The fact that so many links are frozen means the dynamic variables resemble a type of maximal tree gauge, though in this case the non-dynamical variables are not identity. In this picture, for the 4D YM theory, it is no wonder why the integral is so difficult to importance sample. As is well known, using such a maximal tree gauge introduces gross non-locality, and for any long-distance modes in this lattice to turn over, changes need to propagate

through this tree throughout the entire 5D GTM lattice. The tree does presumably contain some overall closed loop structure because of the periodic boundary conditions.

The challenge is to solve this 5D YM system as a function of the fixed links, certainly a challenging prospect. In the limit $M \rightarrow \infty$, the integral can potentially be pictured as this theory with a very weak coupling, however I am not sure how systems behaves with frozen links.

F Derivative operators

A gauge transformation at a single site x will entail updating some links by a left-multiplication and some by a right-multiplication. Such updates are most naturally expressed in terms of a basis of right- and left-invariant differential operators, respectively. In the following $a = 1, \dots, 8$ is an $SU(3)$ adjoint representation index. Right-invariant differential operators $\partial_{x,\mu}^{R,a}$ are defined as

$$\partial_{x,\mu}^{R,a} f(U) = \left. \frac{d}{dt} f(U_t) \right|_{t=0}, \quad U_t(y, \nu) = \begin{cases} e^{tT^a} U(x, \mu) & \text{if } (y, \nu) = (x, \mu), \\ U(y, \nu) & \text{otherwise.} \end{cases} \quad (105)$$

Likewise, left-invariant operators are defined

$$\partial_{x,\mu}^{L,a} f(U) = \left. \frac{d}{dt} f(U_t) \right|_{t=0}, \quad U_t(y, \nu) = \begin{cases} U(x, \mu) e^{-tT^a} & \text{if } (y, \nu) = (x, \mu), \\ U(y, \nu) & \text{otherwise.} \end{cases} \quad (106)$$

The relative minus sign in these definitions is required so that both differential operators satisfy the commutation relations $[\partial^a, \partial^b] = f^{abc} \partial^c$ with the same convention for the structure constants f^{abc} .

G Derivative of adjoint actions

$$\left(e^{\text{adj}(A)} \right)^{ac} T^c = e^A T^a e^{-A} \quad (107)$$

$$d \left(e^{\text{adj}(A)} \right)^{ac} T^c = d \left(e^A T^a e^{-A} \right) \quad (108)$$

$$\begin{aligned} d \left(e^A T^a e^{-A} \right) &= d e^A T^c e^{-A} + e^A T^c d e^{-A} \\ &= d e^A e^{-A} e^A T^c e^{-A} + e^A T^c e^{-A} e^A d e^{-A} \\ &= J(A) A_* e^A T^c e^{-A} - e^A T^c e^{-A} J(-A) A_* \\ &= \{ J(A) \cdot A_* \}^l T^l \left(e^{\text{adj}(A)} \right)^{ac} T^c - \left(e^{\text{adj}(A)} \right)^{ac} T^c \{ J(-A) \cdot A_* \}^l T^l \\ &= \left(e^{\text{adj}(A)} \right)^{ac} \left[\{ J(A) \cdot A_* \}^l T^l T^c - T^c \{ J(-A) \cdot A_* \}^l T^l \right] \end{aligned} \quad (109)$$

H Revision history

1. Created Jan 14, 2025
2. Jan 28, 2025: Adding gauge-fixing field transformation idea.

3. Jan 31, 2025: Norman pointed out in our 01/29/25 group meeting that the naive choice of flow $\partial_{x,\mu}^a \text{ReTr}U(x, \mu)$ is totally local and as such won't provide a correct gauge fixing. Having thought about it more carefully I realized that I agree and that a more appropriate flow is in order. Adding new gauge-fixing flow which is more effective.
4. Feb 3, 2025: Having thought about it a bit more, it may be the case that we need a simultaneous flow along $\partial_{x,\mu}^a \text{ReTr}U(x, \mu)$ and the gauge-fixing force. This is because if we just flow to a configuration which is Landau gauge, there may not be anything to suppresses fluctuations from accumulating in the gauge-direction in the V field space, and so after the first few MD time units the GFFA will lose effectiveness. The issue is that a flow along $\partial_{x,\mu}^a \text{ReTr}U(x, \mu)$ will change the quadratic part of the action $S(U(V))$, and so the action in V space $S(U(V))$ may not be exactly the inverse of the kinetic term of GFFA Hamiltonian **This issue is averted, see Section 4.5.**
5. Feb 6, 2025: Adding description of Peter's perceived issues with gauge fixing flow and possible resolution.
6. Feb 17, 2025: Added Section 4.5 describing partial calculation of the determinant arising from the flow.
7. Feb 27, 2025: Fixed determinant calculation, still have anomalous minus sign in right application determinant.
8. March 5, 2025: Fixed sign of flow, updated flow direction, and added note about not having to propagate forces from Wilson action. Discussion about HMC.
9. March 10, 2025: Added Section 4.4 describing the inverse of the flow.
10. March 21, 2025: Added sentences supporting the reversibility of the flow and Section 4.6 which calculates the Jacobian in terms of the mixed left-/right-invariant basis (Which might even be correct!).

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